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(a)

let w_f be the final value of w
let $t \rightarrow \infty$, so that $w_{t+1} = w_t = w_f$
Therefore: $w_f = (1 - 2\eta\delta^2)w_f + 2\eta\delta y$
We can get $w_f = \frac{y}{\delta}$
let $e_t = w_t - w_f$, $e_{t+1} = w_{t+1} - w_f$
We can get $w_t = w_f + e_t$, $w_{t+1} = w_f + e_{t+1}$
Therefore: $w_f + e_{t+1} = (1 - 2\eta\delta^2)(w_f + e_t) + 2\eta\delta y$
 $\therefore w_f = \frac{y}{\delta}$
 $\therefore \frac{y}{\delta} + e_{t+1} = (1 - 2\eta\delta^2)(\frac{y}{\delta} + e_t) + 2\eta\delta y$
 $\frac{y}{\delta} + e_{t+1} = \frac{y}{\delta} - 2\eta\delta y + (1 - 2\eta\delta^2)e_t + 2\eta\delta y$
 $\therefore e_{t+1} = (1 - 2\eta\delta^2)e_t$
Similarly: $e_t = (1 - 2\eta\delta^2)e_{t-1}, \dots, e_1 = (1 - 2\eta\delta^2)e_0$
Therefore: $e_{t+1} = (1 - 2\eta\delta^2)^{t+1}e_0$
 $\therefore \lim_{t \rightarrow \infty} e_{t+1} = 0$
 $\therefore |1 - 2\eta\delta^2| < 1$
 $-1 < 1 - 2\eta\delta^2 < 1$
 $0 < 2\eta\delta^2 < 2$
 $0 < \eta < \frac{1}{\delta^2}$
In conclusion: $0 < \eta < \frac{1}{\delta^2}$

(b)

I have already proved in (a) that $e_t = (1 - 2\eta\delta^2)^t e_0$

$$\therefore |e_t| = |w_t - w_f| < |w_f - w_f \cdot (1 - \varepsilon)|$$

$$\therefore |1 - 2\eta\delta^2|^t \cdot |e_0| < \varepsilon |w_f|$$

$$\because w_0 = 0$$

$$\therefore |e_0| = |w_0 - w_f| = |w_f|$$

$$\therefore |1 - 2\eta\delta^2|^t |w_f| < \varepsilon |w_f|$$

$$t \cdot \log |1 - 2\eta\delta^2| < \log \varepsilon$$

$$\therefore |1 - 2\eta\delta^2| < 1$$

$$\therefore \log |1 - 2\eta\delta^2| < 0$$

$$\text{Therefore, } t > \frac{\log \varepsilon}{\log |1 - 2\eta\delta^2|}$$

(c)

I have already proved in (a) that $0 < \eta < \frac{1}{\delta^2}$

$$\therefore \delta_l \gg \delta_s$$

$$\therefore \frac{1}{\delta_l^2} < \frac{1}{\delta_s^2}$$

$$\therefore 0 < \eta < \frac{1}{\delta_l^2}$$

(d)

I have already proved in b that $t > \frac{\log \varepsilon}{\log |1-2\eta\delta^2|}$
 $\therefore t_l = \frac{\log \varepsilon}{\log |1-2\eta\delta_l^2|}$, $t_s = \frac{\log \varepsilon}{\log |1-2\eta\delta_s^2|}$

$$\therefore \delta_l^2 \gg \delta_s^2$$

$$\therefore \eta < \frac{1}{\delta_l^2} \ll \frac{1}{\delta_s^2}$$

So, it can be determined that $1-2\eta\delta_s^2 > 0$

$$(i): |1-2\eta\delta_l^2| > |1-2\eta\delta_s^2|$$

If $1-2\eta\delta_l^2 > 0$, we can get $\delta_l^2 < \delta_s^2$, it is impossible

$$\text{Therefore, } 1-2\eta\delta_l^2 < 0 \Rightarrow \eta > \frac{1}{2\delta_l^2}$$

$$2\eta\delta_l^2 - 1 > 1-2\eta\delta_s^2 \Rightarrow \eta > \frac{1}{\delta_l^2 + \delta_s^2}$$

$$\therefore \delta_l^2 \gg \delta_s^2$$

$$\therefore \frac{1}{\delta_l^2 + \delta_s^2} < \eta < \frac{1}{\delta_l^2}$$

$$\therefore \log \varepsilon < 0, \log |1-2\eta\delta_s^2| < \log |1-2\eta\delta_l^2| < 0$$

$$\therefore t_l > t_s$$

$$(ii): |1-2\eta\delta_l^2| < |1-2\eta\delta_s^2|$$

$$\text{that's when } 0 < \eta < \frac{1}{\delta_l^2 + \delta_s^2}$$

$$\therefore \log \varepsilon < 0, \log |1-2\eta\delta_l^2| < \log |1-2\eta\delta_s^2| < 0$$

$$\therefore t_s > t_l$$

$$\text{To conclude: } \begin{cases} t_s < t_l, \frac{1}{\delta_l^2 + \delta_s^2} < \eta < \frac{1}{\delta_l^2} \\ t_s > t_l, 0 < \eta < \frac{1}{\delta_l^2 + \delta_s^2} \end{cases}$$

(e)

I have already proved in (d) that: $\begin{cases} t_s < t_l, & \frac{1}{\sigma_i^2 + \sigma_s^2} < \eta < \frac{1}{\sigma_i^2} \\ t_s > t_l, & 0 < \eta < \frac{1}{\sigma_i^2 + \sigma_s^2} \end{cases}$

(i): when $\frac{1}{\sigma_i^2 + \sigma_s^2} \leq \eta < \frac{1}{\sigma_i^2}$, the speed depends on t_l

$$\therefore 1 - 2\eta\sigma_i^2 < 0$$

$\therefore$ if η get larger, $1 - 2\eta\sigma_i^2$ get smaller but $|1 - 2\eta\sigma_i^2|$ get larger

$\log|1 - 2\eta\sigma_i^2|$ will also get larger but less than 0

So, if η get larger, $\frac{\log \varepsilon}{\log|1 - 2\eta\sigma_i^2|}$ get larger

when $\eta = \frac{1}{\sigma_i^2 + \sigma_s^2}$, t is the smallest

(ii): when $0 < \eta \leq \frac{1}{\sigma_i^2 + \sigma_s^2}$

Similarly, when $\eta = \frac{1}{\sigma_i^2 + \sigma_s^2}$, t is the smallest

To conclude, when $\eta = \frac{1}{\sigma_i^2 + \sigma_s^2}$, we can get the fastest speed

(f)

$\therefore \sigma_l$ is the largest among σ_i

$\therefore \frac{1}{\sigma_l^2}$ is the smallest among the $\frac{1}{\sigma_i^2}$

$\therefore$ They won't influence the choice of possible learning rates

$\therefore$ the speed of learning rate depends on $|1 - 2\eta\sigma_i^2|$



From the graph, we can conclude that the maximum value must occur at one of the two endpoints.

$\therefore \sigma_s$ and σ_l are the endpoints

$\therefore$ Other σ_i won't influence the fastest convergence

(g)

Through singular value decomposition, we can transform the original complicated matrix problem into many independent dimensions, which makes the problem much easier to analyze and solve.

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(a)

In hw00, I have proved that $\text{range}(X^T) = \{X^T a \mid a \in \mathbb{R}^n\}$

$$\therefore r_t = y - Xw_t \in \mathbb{R}^n$$

$$\therefore X^T r_t \in \text{range}(X^T)$$

$$\therefore 2\eta \cdot X^T r_t \in \text{range}(X^T)$$

$$\therefore w_{t+1} = w_t + 2\eta X^T r_t$$

$$w_t = w_{t-1} + 2\eta X^T r_{t-1}$$

$$w_{t-1} = w_{t-2} + 2\eta X^T r_{t-2}$$

$$\vdots$$

$$w_1 = w_0 + 2\eta X^T r_0$$

$$\therefore w_t = w_0 + 2\eta X^T (r_0 + \dots + r_{t-1})$$

$$\therefore \sum_{i=0}^{t-1} r_i \in \mathbb{R}^n$$

$$\therefore 2\eta X^T \sum_{i=0}^{t-1} r_i \in \text{range}(X^T)$$

So that $w_t \in w_0 + \text{range}(X^T)$

Since $w_t - w_0 \in \text{range}(X^T)$, the change from w_0 to w_t has no component in $\text{Null}(X)$, therefore, the null-space component remains unchanged.

(b)

Since X has rank n , $\Sigma \in \mathbb{R}^{n \times d}$ only contains n singulars, and they are all distributed along the diagonal

$$\Sigma = \begin{pmatrix} \sigma_1 & & & 0 & \dots & 0 \\ & \ddots & & & & \\ & & \sigma_n & & & \\ & & & 0 & \dots & 0 \end{pmatrix}_{n \times d} \quad V^T = \begin{pmatrix} V_1^T \\ V_2^T \\ \vdots \\ V_d^T \end{pmatrix} \Rightarrow \Sigma V^T = \begin{pmatrix} \sigma_1 V_1^T \\ \sigma_2 V_2^T \\ \vdots \\ \sigma_n V_n^T \end{pmatrix}$$

We can see that only first n V_i^T in ΣV^T , and $w_t = w_0 + 2\eta U^T (\Sigma V^T) \cdot (r_0 + \dots + r_{t-1})$. Therefore, $\{V_{n+1}, V_{n+2}, \dots, V_d\}$ spans $\text{null}(X)$, n coordinates $V_i^T w_t$ can change, and $(d-n)$ are fixed by w_0 .

(c)

$$W - W_0 = (W - W_{00}) + (W_{00} - W_0)$$

Assuming that there are two values W_{001} and W_{002} fit the solution

$$\therefore XW_{001} = y, XW_{002} = y$$

$$\therefore X(W_{001} - W_{002}) = 0$$

Therefore, $W_{001} - W_{002} \in \text{null}(X)$

$$\therefore W_{001} = W_0 + \text{range}(X^T), W_{002} = W_0 + \text{range}(X^T)$$

$$\therefore W_{001} - W_{002} = (W_0 - W_0) + \text{range}(X^T) \\ = \text{range}(X)$$

However, this is contrary to the conclusion that $W_{001} - W_{002} \in \text{null}(X)$

So, the assumption doesn't hold, W_{00} is unique.

(d)

(1): When $W_0 = 0$, we have to calculate $\min_w \|w\|_2^2$ subject to $Xw = y$

$$\therefore W_t = W_0 + 2\eta X^T(r_0 + \dots + r_{t-1}) \text{ and } W_0 = 0$$

$$\therefore W_t \in \text{range}(X^T)$$

$$\text{So, } W_{00} \in \text{range}(X^T)$$

$$\text{let } W_{00} = X^T a \text{ (} a \in \mathbb{R}^n \text{)}$$

$$y = XW_{00} = XX^T a$$

$$\text{Since } \text{rank}(X) = n$$

$$\therefore XX^T \in \mathbb{R}^{n \times n}$$

$$\therefore XX^T \text{ is invertible}$$

$$\therefore a = (XX^T)^{-1}y$$

Therefore, $W_{00} = X^T a = X^T (XX^T)^{-1}y$, this is the minimum-norm solution

(2): If $W_0 \neq 0$, gradient descent converges to the solution closest to W_0 rather than the one closest to 0

$$\therefore Y = XW + \sqrt{\lambda} N$$

$$\begin{aligned}\therefore E[YY^T] &= E[(XW + \sqrt{\lambda} N)(XW + \sqrt{\lambda} N)^T] \\ &= E[XWW^T X^T + \sqrt{\lambda} XWN^T + \sqrt{\lambda} NW^T X^T + \lambda NN^T] \\ &= E[(XWW^T) \cdot X^T] + \sqrt{\lambda} \cdot E[XWN^T]\end{aligned}$$

$$\therefore W \sim N(0, I)$$

$$\begin{aligned}\therefore E[W] &= 0, \text{cov}(W, W) = E[(W - E[W])(W - E[W])^T] \\ &= I = E[WW^T]\end{aligned}$$

Since W and N are independent, and $E[W] = 0$, $E[N] = 0$

$$\begin{aligned}\text{So, } E(WN^T) &= E[(W - E[W])(N - E[N])^T] \\ &= \text{cov}(W, N) \\ &= 0\end{aligned}$$

$$\therefore E[YY^T] = IX^T + 0 = X^T$$

Therefore, $\sum WY = X^T$

$$\begin{aligned}E[YY^T] &= E[(XW + \sqrt{\lambda} N)(XW + \sqrt{\lambda} N)^T] \\ &= E[XWW^T X^T + \sqrt{\lambda} XWN^T + \sqrt{\lambda} NW^T X^T + \lambda NN^T] \\ &= XE[WW^T]X^T + \sqrt{\lambda} X \cdot E[WN^T] + \sqrt{\lambda} E[NW^T]X^T + \lambda E[NN^T]\end{aligned}$$

$$\therefore N \sim (0, I)$$

$$\begin{aligned}\therefore \text{cov}(N, N) &= E[(N - E[N])(N - E[N])^T] \\ &= I = E[NN^T]\end{aligned}$$

$$\begin{aligned}\therefore E[YY^T] &= X \cdot I \cdot X^T + \sqrt{\lambda} \cdot X \cdot 0 + \sqrt{\lambda} \cdot 0 \cdot X^T + \lambda \cdot I \\ &= XX^T + \lambda I\end{aligned}$$

Therefore, $\sum Y Y = XX^T + \lambda I$

$$\therefore \sum WY = X^T, \sum Y Y = XX^T + \lambda I$$

$$\therefore E[W|Y=y] = X^T(XX^T + \lambda I)^{-1}y$$

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(a)

Substituting $X = U\Sigma V^T$ into $W_{t+1} = W_t + 2\eta X^T(y - XW_t)$:

$$\begin{aligned} W_{t+1} &= W_t + 2\eta V\Sigma^T U^T (y - U\Sigma V^T W_t) \\ &= W_t + 2\eta V\Sigma^T U^T y - 2\eta V\Sigma^T U^T U \Sigma (V^T W_t) \end{aligned}$$

$\therefore U$ and V are orthogonal matrix

$$\therefore U^T U = I, \quad V^T V = I$$

$$W_{t+1} = W_t + 2\eta V\Sigma^T (U^T y) - 2\eta V\Sigma^T \Sigma (V^T W_t)$$

$$V^T W_{t+1} = V^T W_t + 2\eta (V^T V) \Sigma^T (U^T y) - 2\eta (V^T V) \Sigma^T \Sigma (V^T W_t)$$

$$\tilde{W}_{t+1} = \tilde{W}_t + 2\eta \Sigma^T \tilde{y} - 2\eta \Sigma^T \Sigma \tilde{W}_t$$

$$\therefore \Sigma^T = \begin{pmatrix} \sigma_1 & & & 0 & \dots & 0 \\ & \ddots & & & & \\ & & \sigma_r & & & \\ & & & 0 & \dots & 0 \\ & & & & \ddots & \\ & & & & & 0 \end{pmatrix}_{d \times n}, \quad \Sigma^T \Sigma = \begin{pmatrix} \sigma_1^2 & & & 0 & \dots & 0 \\ & \ddots & & & & \\ & & \sigma_r^2 & & & \\ & & & 0 & \dots & 0 \\ & & & & \ddots & \\ & & & & & 0 \end{pmatrix}_{d \times d}$$

$$\therefore \Sigma^T \tilde{y} = \begin{pmatrix} \sigma_1 \tilde{y}_1 \\ \sigma_1 \tilde{y}_1 \\ \sigma_r \tilde{y}_r \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \Sigma^T \Sigma \tilde{W}_t = \begin{pmatrix} \sigma_1^2 \tilde{w}_1 \\ \sigma_1^2 \tilde{w}_1 \\ \sigma_r^2 \tilde{w}_r \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$\text{Therefore, if } i \leq r: [\tilde{W}_{t+1}]_i = [\tilde{W}_t]_i + 2\eta \sigma_i \tilde{y}_i - 2\eta \sigma_i^2 [\tilde{W}_t]_i$$

$$\text{if } r < i \leq d: [\tilde{W}_{t+1}]_i = [\tilde{W}_t]_i + 0 - 0$$

$$\text{To conclude, } \begin{cases} [\tilde{W}_{t+1}]_i = (1 - 2\eta \sigma_i^2) [\tilde{W}_t]_i + 2\eta \sigma_i \tilde{y}_i \\ [\tilde{W}_{t+1}]_i = [\tilde{W}_t]_i \end{cases}$$

(b)

I have already proved in a that while $i \leq r: [\tilde{w}_{t+1}]_i = (1-2\eta\delta_i^2)[\tilde{w}_t]_i + 2\eta\delta_i^2\tilde{y}_i$

Let $[\tilde{w}_{t+1}]_i - \beta = (1-2\eta\delta_i^2)([\tilde{w}_t]_i - \beta)$

$$\begin{aligned} [\tilde{w}_{t+1}]_i &= (1-2\eta\delta_i^2)[\tilde{w}_t]_i + \beta - \beta + 2\eta\delta_i^2\beta \\ &= (1-2\eta\delta_i^2)[\tilde{w}_t]_i + 2\eta\delta_i^2\beta \end{aligned}$$

$$\therefore 2\eta\delta_i^2\beta = 2\eta\delta_i^2\tilde{y}_i$$

$$\beta = \frac{\tilde{y}_i}{\delta_i}$$

$$\text{Therefore, } [\tilde{w}_{t+1}]_i - \frac{\tilde{y}_i}{\delta_i} = (1-2\eta\delta_i^2)[[\tilde{w}_t]_i - \frac{\tilde{y}_i}{\delta_i}]$$

$$\because w_0 = 0$$

$$\therefore [\tilde{w}_t]_i - \frac{\tilde{y}_i}{\delta_i} = (1-2\eta\delta_i^2)^t (0 - \frac{\tilde{y}_i}{\delta_i})$$

$$[\tilde{w}_t]_i = \frac{\tilde{y}_i}{\delta_i} [1 - (1-2\eta\delta_i^2)^t]$$

$$\text{While } i > r, [\tilde{w}_{t+1}]_i = [\tilde{w}_t]_i$$

$$\therefore [\tilde{w}_t]_i = [\tilde{w}_0]_i = 0$$

$$\text{Therefore } [\tilde{w}_t] = \begin{cases} \frac{\tilde{y}_i}{\delta_i} [1 - (1-2\eta\delta_i^2)^t], & i \leq r \\ 0, & i > r \end{cases}$$

$$\because 0 < 2\eta\delta_i^2 < 1$$

$$\therefore 0 < 1 - 2\eta\delta_i^2 < 1$$

$$\therefore 0 < (1-2\eta\delta_i^2)^t < 1, (1-2\eta\delta_i^2)^t \text{ decrease when } t \text{ increase}$$

$$\therefore 0 < 1 - (1-2\eta\delta_i^2)^t < 1$$

$$\therefore S_i^{GD}(t) \in (0, 1) \text{ and } S_i^{GD}(t) \text{ increase when } t \text{ increase}$$

$$\therefore \lim_{t \rightarrow \infty} (1-2\eta\delta_i^2)^t = 0$$

$$\therefore \lim_{t \rightarrow \infty} 1 - (1-2\eta\delta_i^2)^t = 1$$

$$\therefore \lim_{t \rightarrow \infty} S_i^{GD}(t) = 1$$

(c)

(1): $\because 1 - x \leq e^{-x}$,

Substituting $x = 2\eta\sigma_i^2$ into $1 - x \leq e^{-x}$:

We can get $1 - 2\eta\sigma_i^2 \leq e^{-2\eta\sigma_i^2}$

$\therefore (1 - 2\eta\sigma_i^2)^t \leq e^{-2\eta t\sigma_i^2}$

$\therefore 1 - (1 - 2\eta\sigma_i^2)^t \geq 1 - e^{-2\eta t\sigma_i^2}$

So that $S_i^{GD}(t) \geq 1 - e^{-2\eta t\sigma_i^2}$

(2): when $\lambda = \frac{1}{2\eta t}$, $\sigma_i^2 = \lambda$

$S_i^{ridge}(\lambda) = \frac{\sigma_i^2}{\sigma_i^2 + \lambda} = \frac{\lambda}{2\lambda} = \frac{1}{2}$

$S_i^{GD}(t) = 1 - (1 - 2\eta \cdot \lambda)^t$
 $= 1 - (1 - 2\eta \cdot \frac{1}{2\eta t})^t$
 $= 1 - (1 - \frac{1}{t})^t$

when $t \geq 1$, $0 < (1 - \frac{1}{t})^t < 1$

$\therefore 0 < 1 - (1 - \frac{1}{t})^t < 1$

$\therefore$ when $\lambda = \frac{1}{2\eta t}$, $S_i^{ridge}(\lambda)$ and $S_i^{GD}(t)$ are of the same order of magnitude

Therefore, when $\lambda = \frac{1}{2\eta t}$, running gradient descent for t steps acts like ridge regression.

As t grows, λ decrease, and it approaches 0 as $t \rightarrow \infty$

(d)

$$S_i^{GD}(t) = 1 - (1 - 2\eta\sigma_i^2)^t$$

When $2\eta\sigma_i^2 > 1$, $1 - 2\eta\sigma_i^2 < 0$

When t is odd, $(1 - 2\eta\sigma_i^2)^t < 0$

$$\therefore 1 - (1 - 2\eta\sigma_i^2)^t > 1$$

Therefore, if $2\eta\sigma_i^2 > 1$, when t is odd, $S_i^{GD}(t) > 1$

Let's come to ridge regression:

$$\text{When } \lambda > 0, S_i^{\text{ridge}}(\lambda) = \frac{\sigma_i^2}{\sigma_i^2 + \lambda} < \frac{\sigma_i^2}{\sigma_i^2} = 1$$

$$\therefore [V^T W_{\text{ridge}}(\lambda)]_i = S_i^{\text{ridge}}(\lambda) \cdot \frac{\tilde{y}_i}{\sigma_i} < \frac{\tilde{y}_i}{\sigma_i}$$

Therefore, if $\lambda > 0$, this will not happen

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(a)

The training datas contain noise, while the test datas are noiseless.

(b)

As the hidden layer width increases, the network has more ReLU units and therefore more possible elbows. This makes the learned function more flexible and allows it to better approximate the piecewise-linear target, which generally reduces the test error.

I would place them near the locations where the target function changes slope.

(c)

Ridge regression is the most computationally efficient because it directly solves for the output-layer weights instead of using many SGD iterations. Training only the output layer with SGD is slower, while training all layers is generally the most expensive. The ReLU elbows remain fixed when only the output layer is trained, either with SGD or ridge regression, because the hidden-layer parameters do not change. When all layers are trained, the elbows can move. This is important because the network can move the slope-change locations toward the bends in the target function, which can improve the learned function and reduce test error.